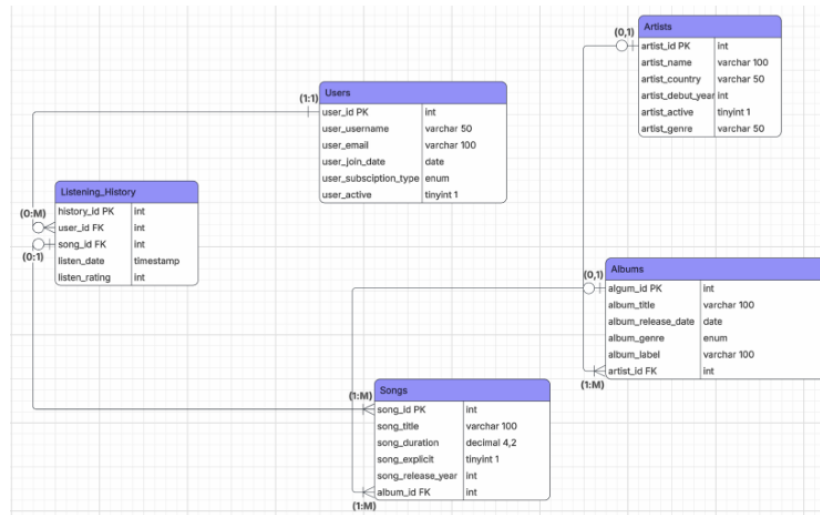


Project Elements:

1. Please include the names of all the members of your design team
 - Clayton Glazier
 - Michael Bell and Pedro Calmon-Coelho were my partners on the previous SQL project
2. An image of your old SQL ERD showing the original relationship between the entities you are working with



Part 1 Design

3. Design 1 (16 points)

- **3.1 Screenshot(s) showing Design 1** implemented in Mongo. Be sure to include sufficient screenshots to show the complete implemented design and data set.



```

{
  "_id": "artist_drake",
  "artist_name": "Drake",
  "artist_country": "Canada",
  "artist_debut_year": 2006,
  "artist_active": true,
  "artist_genre": "Hip-Hop",
  "album_ids": [
    "Views",
    "Take_Care"
  ]
}

{
  "_id": "artist_tswift",
  "artist_name": "Taylor Swift",
  "artist_country": "USA",
  "artist_debut_year": 2006,
  "artist_active": true,
  "artist_genre": "Pop",
  "album_ids": [
    "1989",
    "Folklore"
  ]
}

```

- **3.2. Explanation For Design 1 “array”** Explain how the relationship is 1:M. Explain which is the 1 side and which is the M side. Explain how the values in the array are limited and thus it would not create an infinite array. What is the limit? (note, if there is no limit, then you have chosen the wrong example!)

This is a 1:M relationship because one artist has many albums. The artist is the 1 side, and the albums listed in the album_ids array are the M side. The array is naturally limited because an artist can only release a finite number of albums, so the list cannot grow indefinitely.

- **3.3 JSON documents for Design 1.** Provide a JSON document showing design 1. Merely showing the records in Mongo is insufficient. You must provide a separate .json or text file that I can run through the validator.

4. Design 2 (17 points):

- **4.1 Screenshot(s) showing Design 2 implemented in Mongo.** Be sure the subdocuments are expanded so I can see all the elements! Be sure to include sufficient screenshots to show the complete implemented design and data set.

```
{
  "_id": "album_views",
  "album_title": "Views",
  "album_release_year": 2016,
  "album_genre": "Hip-Hop",
  "artist": {
    "artist_id": "artist_drake",
    "artist_name": "Drake",
    "artist_country": "Canada",
    "artist_debut_year": 2006,
    "artist_active": true,
    "artist_genre": "Hip-Hop"
  }
}
```

```
{
  "_id": "album_take_care",
  "album_title": "Take Care",
  "album_release_year": 2011,
  "album_genre": "Hip-Hop",
  "artist": {
    "artist_id": "artist_drake",
    "artist_name": "Drake",
    "artist_country": "Canada",
    "artist_debut_year": 2006,
    "artist_active": true,
    "artist_genre": "Hip-Hop"
  }
}
```

```
{
  "_id": "album_1989",
  "album_title": "1989",
  "album_release_year": 2014,
  "album_genre": "Pop",
  "artist": {
    "artist_id": "artist_tswift",
    "artist_name": "Taylor Swift",
    "artist_country": "USA",
    "artist_debut_year": 2006,
    "artist_active": true,
    "artist_genre": "Pop"
  }
}
```

```
{
  "_id": "album_folklore",
  "album_title": "Folklore",
  "album_release_year": 2020,
  "album_genre": "Pop",
  "artist": {
    "artist_id": "artist_tswift",
    "artist_name": "Taylor Swift",
    "artist_country": "USA",
    "artist_debut_year": 2006,
    "artist_active": true,
    "artist_genre": "Pop"
  }
}
```

- **4.1 Explanation For Design 2 “subdocuments”** Explain which is the 1 side and which is the M side.

This is a 1:M relationship because one artist appears in multiple album documents. The artist is the 1 side, and the albums are the M side.

- **4.2 JSON documents for Design 2.** Provide a JSON document showing design 2. Merely showing the records in Mongo is insufficient. You must provide a separate .json or text file that I can run through the validator

5. Design 3 (17 points):

- 5.1 Screenshot(s) showing Design 3 implemented in Mongo. Be sure the subdocuments are expanded so I can see all the elements! Be sure to include sufficient screenshots to show the complete implemented design and data set.

```
_id: "user_001"
user_username: "musicfan01"
user_email: "musicfan01@example.com"
user_subscription_type: "Premium"
user_active: true
▼ listening_history: Array (2)
  ▼ 0: Object
    history_id: "hist_001"
    song_title: "Views"
    listen_date: "2024-04-01T10:15:00Z"
    listen_rating: 5
  ▼ 1: Object
    history_id: "hist_002"
    song_title: "1989"
    listen_date: "2024-04-02T21:30:00Z"
    listen_rating: 4
```

```
_id: "user_002"
user_username: "swiftie_fan"
user_email: "swiftie@example.com"
user_subscription_type: "Free"
user_active: true
▼ listening_history: Array (2)
  ▼ 0: Object
    history_id: "hist_003"
    song_title: "Folklore"
    listen_date: "2024-03-15T08:00:00Z"
    listen_rating: 5
  ▼ 1: Object
    history_id: "hist_004"
    song_title: "Take_Care"
    listen_date: "2024-03-16T19:45:00Z"
    listen_rating: 3
```

- 5.2 Explanation for Design 3 “subdocuments in an array” Explain which is the 1 side and which is the M side.

This is a 1:M relationship because one user has many listening history entries. The user is the 1 side, and the listening history subdocuments stored in the array are the M side. Each listening event is stored as its own subdocument inside the user document.

- 5.3 JSON documents for Design 3. Provide a JSON document showing design 3. Merely showing the records in Mongo is insufficient. You must provide a separate .json or text file that I can run through the validator

Part 2. Schémas and Queries

1. **At least one example Schema (20 points)** for Design 1, 2 or 3 (choose 1) saved as .json files or text file. Merely showing the schema loaded in Compass is not sufficient, you must provide a separate .json file. Each schema must include:
 - Schema title and description
 - Descriptions for each elements
 - At least 1 required element (beyond _id)
 - Limitations/rules for at least 2 elements
2. **Validation (5 points):** A screenshot that provides proof that your schema validates. For partial credit, a screenshot showing that you tried but could not achieve validation for the schema.

The screenshot shows the MongoDB Compass Schema Designer interface. At the top, a JSON schema is displayed:

```
{  "listening_history": {    "bsonType": "array",    "description": "Array of listening events.",    "items": {      "bsonType": "object"    }  }}
```

 Below the schema, there is an "EXPORT SCHEMA" button. The main part of the interface shows a list of fields for a document. The fields are:

- _id**: string, with example values "user_002" and "user_001".
- listening_history**: array of documents, with a description "Array of documents with 4 nested fields." and statistics: min: 2, average: 2.0, max: 2.
- user_active**: boolean, with a value of "true".
- user_email**: string, with example values "musicfan0@example.com" and "swiftie@example.com".
- user_subscription_type**: string, with example values "Free" and "Premium".
- user_username**: string, with example values "musicfan01" and "swiftie_fan".

3. **Queries(20 points):** Text for the following queries and screenshots showing the results (any design)
 - Greater than or less than query

The screenshot shows the MongoDB Compass interface. At the top, a query is entered:

```
{ 'listening_history.listen_rating': { '$lt': 4 } }
```

 Below the query, there are buttons for "ADD DATA", "UPDATE", "DELETE", and "EXPORT DATA". The results are displayed in a JSON format:

```
{  "_id": "user_002",  "user_username": "swiftie_fan",  "user_email": "swiftie@example.com",  "user_subscription_type": "Free",  "user_active": true,  "listening_history": [    {    "history_id": "hist_003",    "song_title": "Folklore",    "listen_date": "2024-03-15T08:00:00Z",    "listen_rating": 5    },    {    "history_id": "hist_004",    "song_title": "Take_Care",    "listen_date": "2024-03-16T19:45:00Z",    "listen_rating": 3    }  ]}
```

- Query within an array

🕒 ▼ {'listening_history.song_title':'Folklore'}

+ ADD DATA ▼
 UPDATE
 DELETE
 EXPORT DATA

```

_id: "user_002"
user_username: "swiftie_fan"
user_email: "swiftie@example.com"
user_subscription_type: "Free"
user_active: true
▼ listening_history: Array (2)
  ▼ 0: Object
    history_id: "hist_003"
    song_title: "Folklore"
    listen_date: "2024-03-15T08:00:00Z"
    listen_rating: 5
  ▼ 1: Object
    history_id: "hist_004"
    song_title: "Take_Care"
    listen_date: "2024-03-16T19:45:00Z"
    listen_rating: 3
  
```

- Query within a subdocument

🕒 ▼ {'artist.artist_country':'USA'}

+ ADD DATA ▼
 UPDATE
 DELETE
 EXPORT DATA

```

▼ {
  "_id": "album_1989",
  "album_title": "1989",
  "album_release_year": 2014,
  "album_genre": "Pop",
  ▼ "artist": {
    "artist_id": "artist_tswift",
    "artist_name": "Taylor Swift",
    "artist_country": "USA",
    "artist_debut_year": 2006,
    "artist_active": true,
    "artist_genre": "Pop"
  }
}

```

▶ ▼ {

```

  "_id": "album_folklore",
  "album_title": "Folklore",
  "album_release_year": 2020,
  "album_genre": "Pop",
  ▼ "artist": {
    "artist_id": "artist_tswift",
    "artist_name": "Taylor Swift",
    "artist_country": "USA",
    "artist_debut_year": 2006,
    "artist_active": true,
    "artist_genre": "Pop"
  }
}

```